

IMPORTS

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

EXPORT / IMPORT

```
df.to_csv('f.csv', index=False)
pd.read_csv('f.csv')
df.to_excel('f.xlsx', index=False)
pd.read_excel('f.xlsx')
df.to_pickle('f.pickle')
pd.read_pickle('f.pickle')
df.to_parquet('f.parquet')
pd.read_parquet('f.parquet')
```

DEPRECATED / REMOVED

```
.ix[...] -> .loc[...] / .iloc[...]
df.append(...) -> pd.concat([...])
sns.distplot(...) -> sns.histplot() / sns.kdeplot()
df.fillna(method='ffill') -> df.ffill()
df.fillna(method='bfill') -> df.bfill()
inplace=True -> df = df.method(...)
```

LOAD & INSPECT

```
df = pd.read_csv('data.csv')
df = pd.read_csv('data.tsv', sep='\t')
df.head()
df.tail()
df.shape
df.columns
df.dtypes
df.info()
df.describe()
```

MISSING DATA

```
pd.isnull(x)
pd.notnull(x)
pd.isna(x)
pd.notna(x)
df.isnull().sum()
df.fillna(0)
df.ffill()
df.bfill()
df.interpolate()
df.dropna()
df.dropna(how='all')
df.dropna(thresh=N)
df['col'].sum(skipna=True)
```

SUBSETTING

```
df['col']
df[['col1', 'col2']]
df.iloc[:, n]
df.loc[label]
df.iloc[pos]
df.loc[row_sel, col_sel]
df[df['col'] > x]
df[(cond1) & (cond2)]
df.query('cond1 and cond2')
```

TIDY & RESHAPING

```
pd.melt(df, id_vars=['id'], var_name='var',
        value_name='val')
df.pivot_table(index='id', columns='var',
               values='val').reset_index()
df['col'].str.split('_', expand=True)
df[['a','b']].drop_duplicates()
df.merge(sub, on=['a','b'])
glob.glob('data/*.csv')
pd.concat([pd.read_csv(f) for f in files])
```

GROUPBY / AGGREGATION

```
df.groupby('col')['target'].mean()
df.groupby(['col1', 'col2'])[['t1', 't2']].mean()
df.groupby('col').agg({'t1': 'mean', 't2': 'sum'})
df.groupby('col').agg(avg=('t1', 'mean'), total=('t2',
        'sum'))
df['col'].nunique()
df['col'].value_counts()
```

PLOTTING

```
df['col'].plot.hist()
df.plot.scatter(x='a', y='b')
sns.histplot(df['col'])
sns.scatterplot(x='a', y='b', data=df)
sns.boxplot(x='cat', y='num', data=df, hue='cat2')
sns.lmplot(x='a', y='b', data=df, col='facet_col')
fig, ax = plt.subplots()
```

CREATING & MODIFYING

```
pd.Series([1,2,3], index=['a','b','c'])
pd.DataFrame({'col1': [...], 'col2': [...]})
df['new_col'] = ...
df.rename(columns={'old':'new'})
df.drop(columns=['col'])
df['col'].apply(func)
df.apply(lambda row: ..., axis=1)
```

COMBINING DATA

```
pd.concat([df1, df2])
pd.concat([df1, df2], axis=1)
pd.concat([df1, df2], ignore_index=True)
pd.concat([df1, df2], join='inner')
left.merge(right, on='key')
left.merge(right, left_on='a', right_on='b',
        how='outer')
```